Business Intelligence CAPSTONE Project
Progress Report

3DSA-2 Group 1

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Business Overview:

DataCo Global is the focus company in the DataCo Smart Supply Chain dataset (Constante, 2019). The dataset covers provisioning, production, sales, and commercial distribution, focusing on products such as clothing, sports, and electronic supplies. The group used DataCo as a case study company to support their analysis and made no claims about any real business opportunities.

The business operates by:

- **Customers:** consumers, corporate buyers, and home offices who order online.
- **Markets:** orders are placed across Africa, Europe, Latin America, Pacific Asia, United States, and Canada
- **Products:** items are organized by department and category, including apparel, footwear, fitness, outdoor, golf, and technology.
- **Payments:** customers may pay by debit, bank transfer, cash, or online payment.
- **Fulfillment:** orders ship under four modes (Standard Class, Second Class, First Class and Same Day), each with scheduled delivery days. Orders move through statuses such as complete, pending, canceled, and suspected fraud.

DataCo operates globally; therefore, delivery performance, margins, and order quality directly impact customer satisfaction and profit. Sales, shipping, and website data are currently stored separately, making it hard for management to obtain a comprehensive overview of what's happening to the business. Implementing a business intelligence solution that integrates the data sources would provide managers with a unified platform to monitor performance and address issues proactively.

Data Overview:

The data utilized for DataCo is from "DataCo Smart Supply Chain for Big Data Analysis" published on Mendeley Data and Kaggle. The main file used was sales file "DataCoSupplyChainDataset.csv", which contains 180,519 rows and 53 columns with a grain of one row per order item. This also covers orders from January 2015 to early 2018.

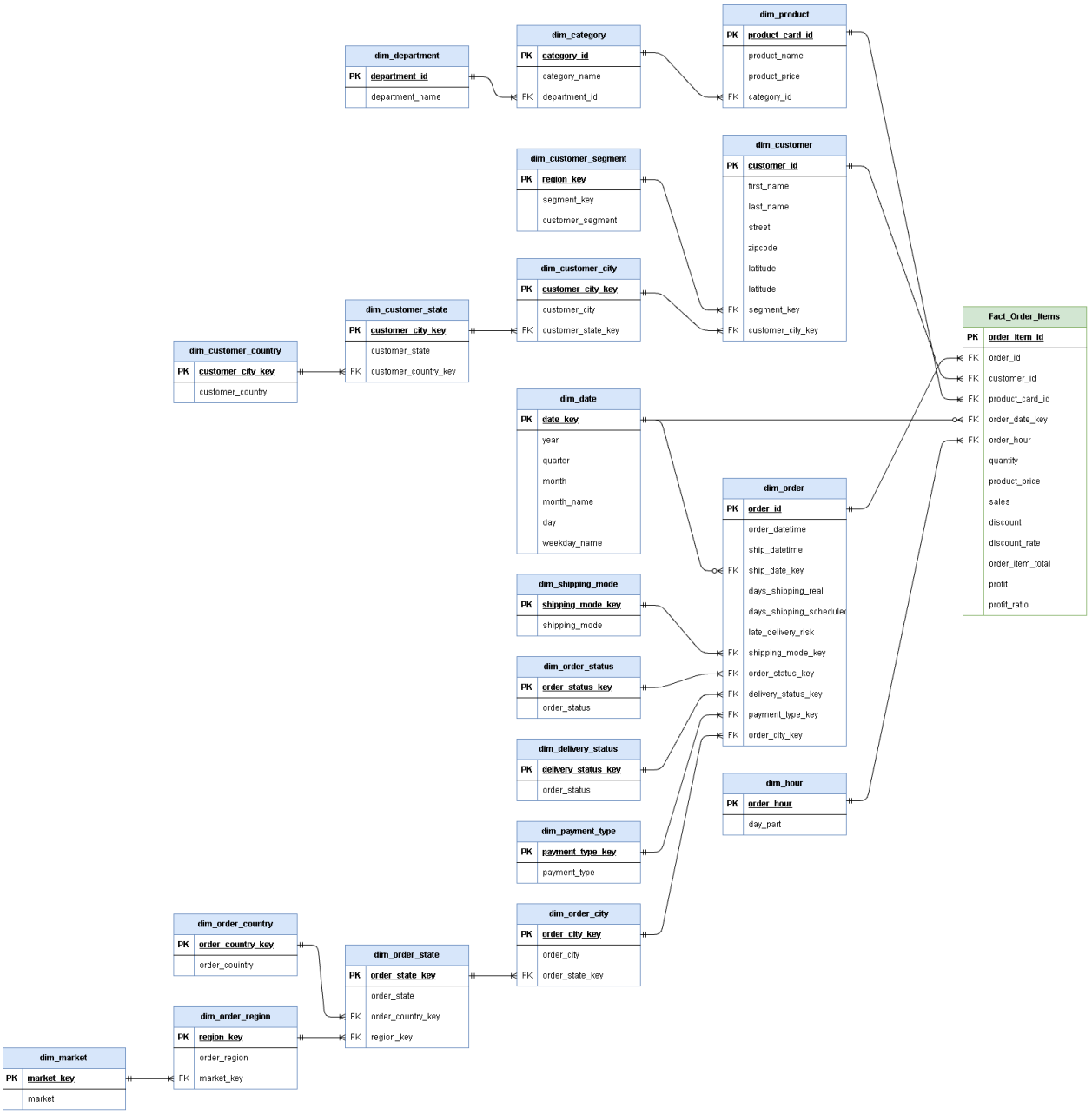
Data Source Schema:

The datasource schema for DataCo contains 21 tables built and cleaned with MySql. The table below shows the different groups, tables, and its specific key columns.

Group	Table	Key Columns
Fact	fact_order_items	order_item_id (PK), order_id, customer_id, product_card_id, order_date_key, order_hour, quantity, product_price, sales, discount, discount_rate, order_item_total, profit, profit_ratio
Order	dim_order	order_id (PK), order_datetime, ship_datetime, days_shipping_real, days_shipping_scheduled,

		late_delivery_risk, shipping_mode_key, order_status_key, delivery_status_key, payment_type_key, order_city_key
	dim_order_status	order_status_key, order_status
	dim_shipping_mode	shipping_mode_key, shipping_mode
	dim_delivery _status	delivery_status_key, delivery_status
	dim_payment_type	payment_type_key, payment_type
Customer	dim_customer	customer_id (PK), first_name, last_name, street, zipcode, latitude, longitude, segment_key, customer_city_key
	dim_customer_segment	segment_key, customer_segment
	dim_customer_country	customer_country_key, customer_country
	dim_customer_state	customer_state_key, customer_state, customer_country_key
	dim_customer_city	customer_city_key, customer_city, customer_state_key
Product	dim_product	product_card_id (PK), product_name, product_price, product_image, category_id
	dim_category	category_id, category_name, department_id
	dim_department	department_id, department_name
Order Destination	dim_order_country	order_country_key, order_country
	dim_order_state	order_state_key, order_state, order_country_key, region_key
	dim_order_region	region_key, order_region, market_key
	dim_order_city	order_city_key, order_city, order_state_key
	dim_market	market_key, market
Time	dim_date	date_key (PK), year_, quarter_, month_, month_name, day_, weekday_name, is_weekend
	dim_hour	hour_key, day_part

Entity Relationship Diagram



Problem Statement:

DataCo Global, an online retailer that sells clothing, sports gear, and electronics around the world. Their sales, shipping, and website data are stored separately, which makes it hard for managers to spot late deliveries, underperforming markets, or the impact of discounts and cancellations on profits. This project will create a data warehouse and an interactive dashboard to bring all this information together and track key performance indicators. With these tools, DataCo's managers will be able to find issues with delivery, profitability, and order quality sooner.

Objectives:

- Clean and build a normalized schema with 20 tables from the sales, shipping, and web log data.
- Track and analyze sales, profit, and discounting by market, department, and customer segment.
- Evaluate delivery performance through shipping mode and market.
- Identify canceled and potentially fraudulent orders for further investigation.

Key Business Questions:

1. How are the monthly sales trending, and which months show growth or decline?
2. Which markets, departments, and customer segments generate great sales and profit?
3. Which departments or products fall below the marginal target?
4. Does adding discounts reduce profit, and by how much?
5. What share of orders is late, how does it vary by shipping mode and market?
6. How different are the actual shipping days from the scheduled days?
7. What orders share the same characteristics of canceled or suspected fraud?
8. Which payment types and markets show higher cancellation or fraud rates?
9. How many customers repeat orders?
10. Do late deliveries reduce repeat orders?

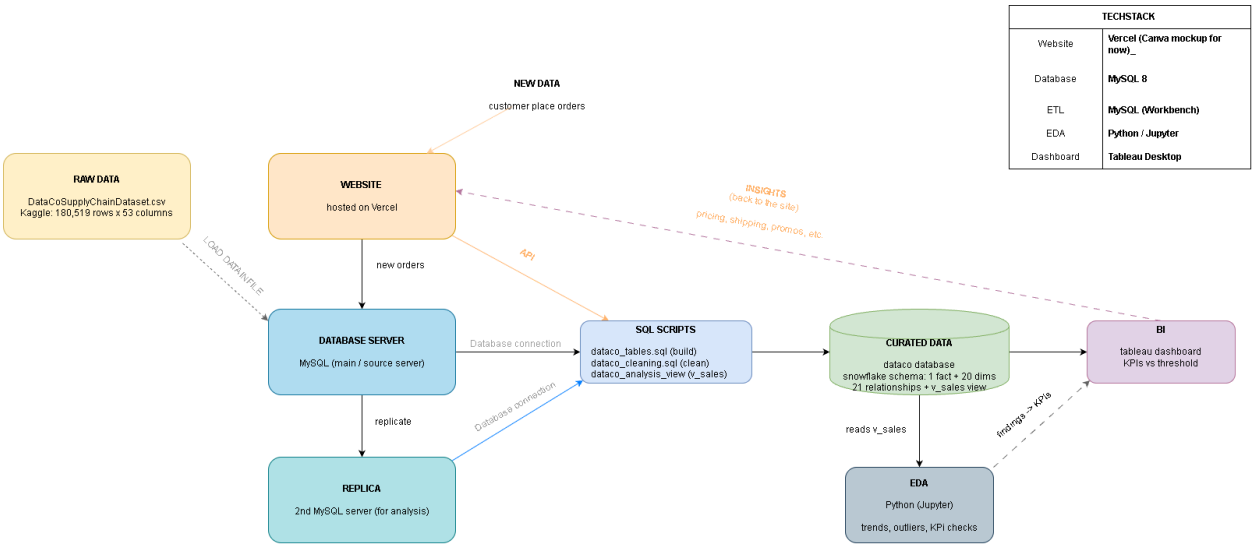
KPI Metrics and Thresholds:

The groups selected seven KPI metrics that would cover the problems stated which are profitability, discounting, delivery, order quality, customer satisfaction, and growth. There are specific thresholds for each KPI to identify if each KPI reaches its target result or it is in a dangerous warning zone.

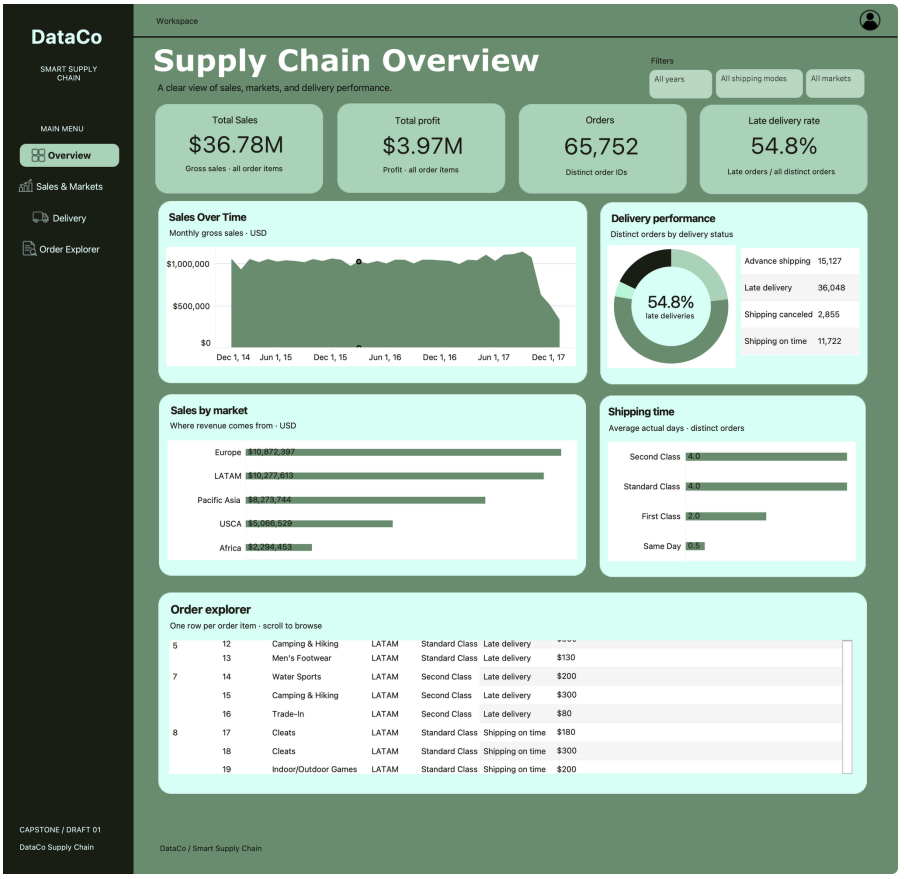
KPI	Formula	Target	Warning	Purpose
Profit Margin	Total Order Profit / Total Sales	12% or higher	Below 5%	Shows whether the business sales are making positive money
Monthly Sales Growth	(Sales of current month - Sales of previous month) / Sales of previous month	3% or higher	Below 0%	Shows the business direction based on sales
Average Discount Rate	Total Discount / Gross Item Price	10% or lower	Above 20%	Shows how discounts affect the profit
Late Delivery Rate	Late Orders / All Orders	15% or lower	Above 30%	Shows how often late order occur from every order
Average Delivery	Average of (Actual - Scheduled Shipping Days)	0.5 days or lower	Above 1.5 days	Shows how the actual delivery

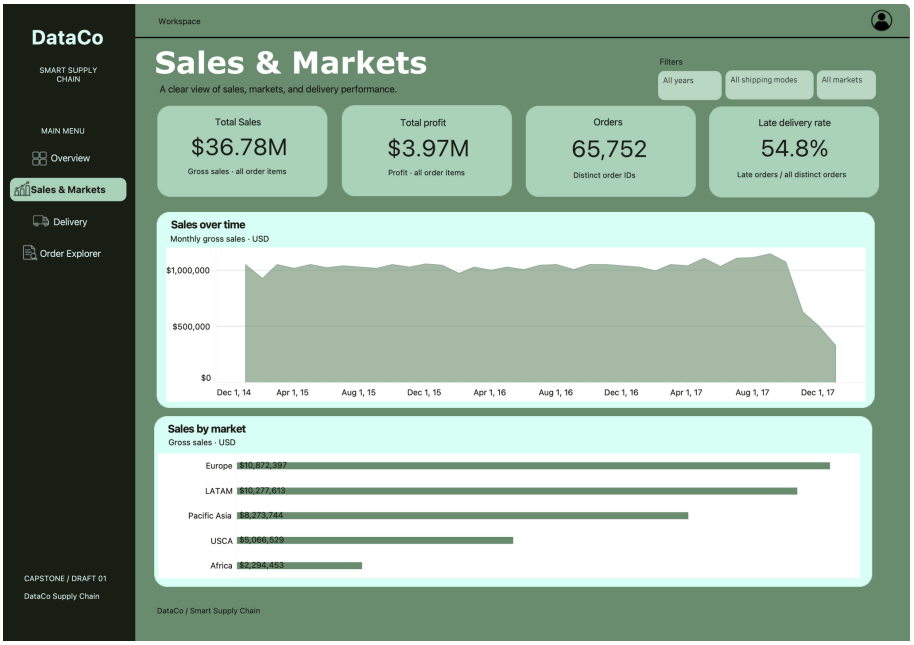
Delay	per order			occurs from the scheduled delivery
Cancelled/Suspected Fraud Rate	Orders with Cancelled/Suspected Fraud Rate status / All Orders	5% or lower	Above 8%	Shows how much of the order quality are deemed cancelled/fraud
Repeat Customer Rate	Customer (2 or more orders) / all customers	30% or higher	Below 15%	Shows customer loyalty

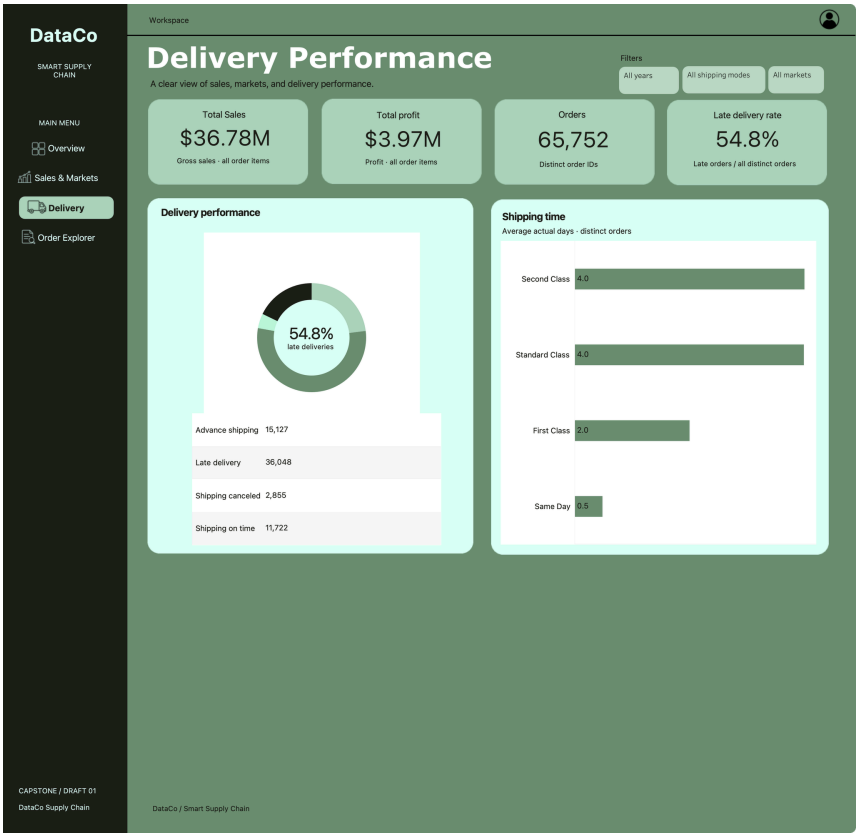
Solution Architecture:

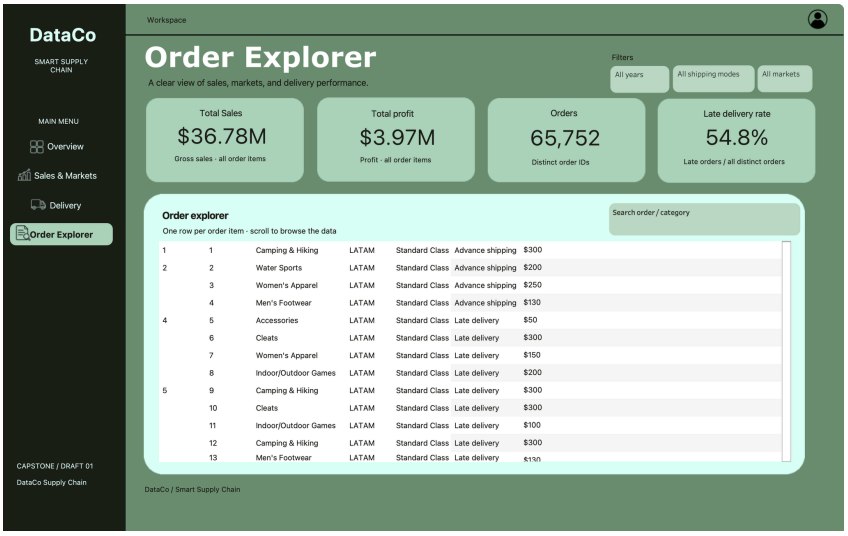


Dashboard Overview:









Website UI/UX Overview:

[DataCo Website]

Slide Deck Link:

[Group1-3DSA2-Slide Deck]

References:

Constante, F. (2019). DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS. *Data Archiving and Networked Services (DANS)*, *5*. https://doi.org/10.17632/8gx2fvg2k6

Tiwari, S. (2019, December 5). *DataCo smart supply chain for big data analysis*. Kaggledatasets. https://www.kaggle.com/datasets/shashwatwork/dataco-smart-supply-chain-for-big-data-analysis

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